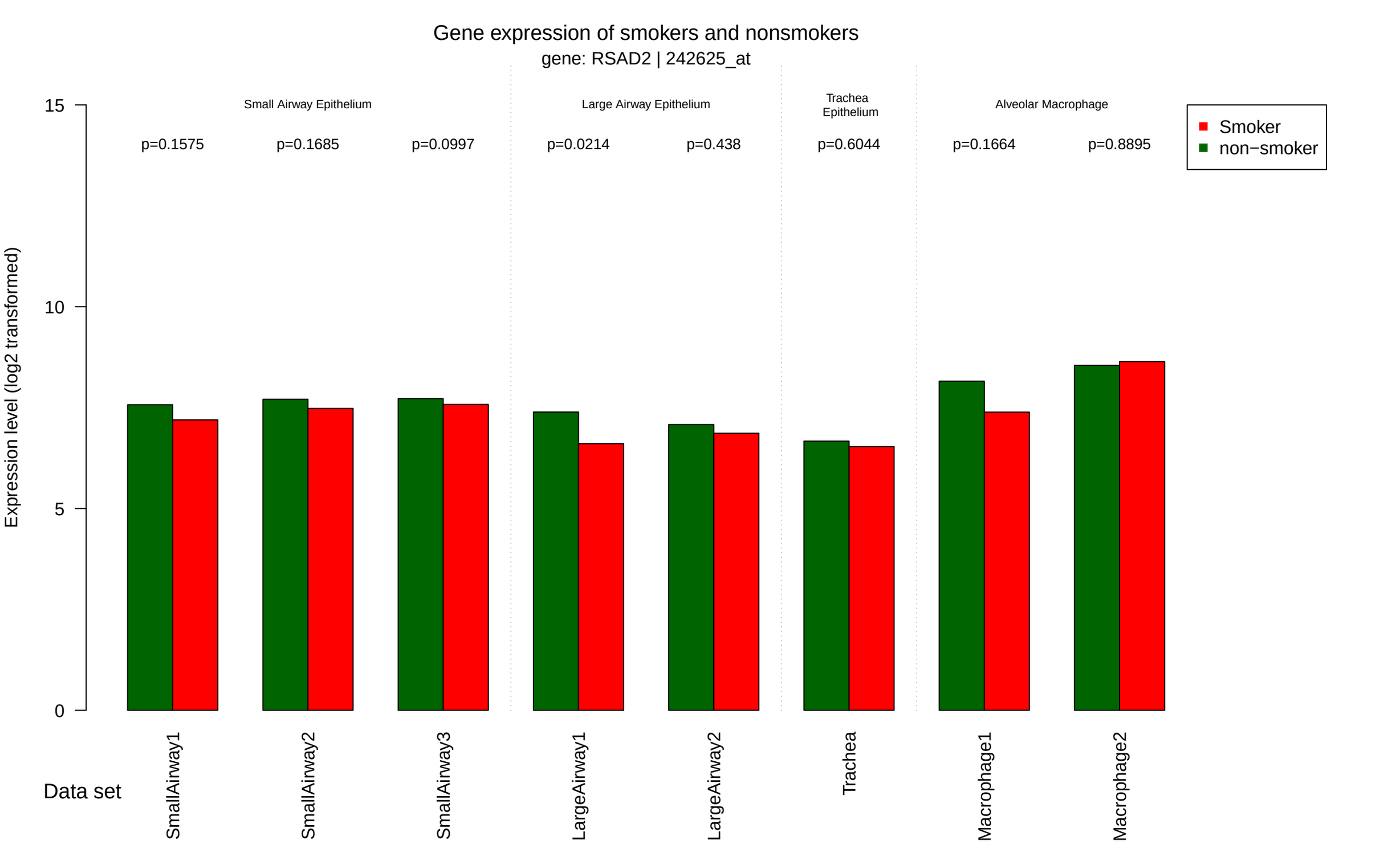
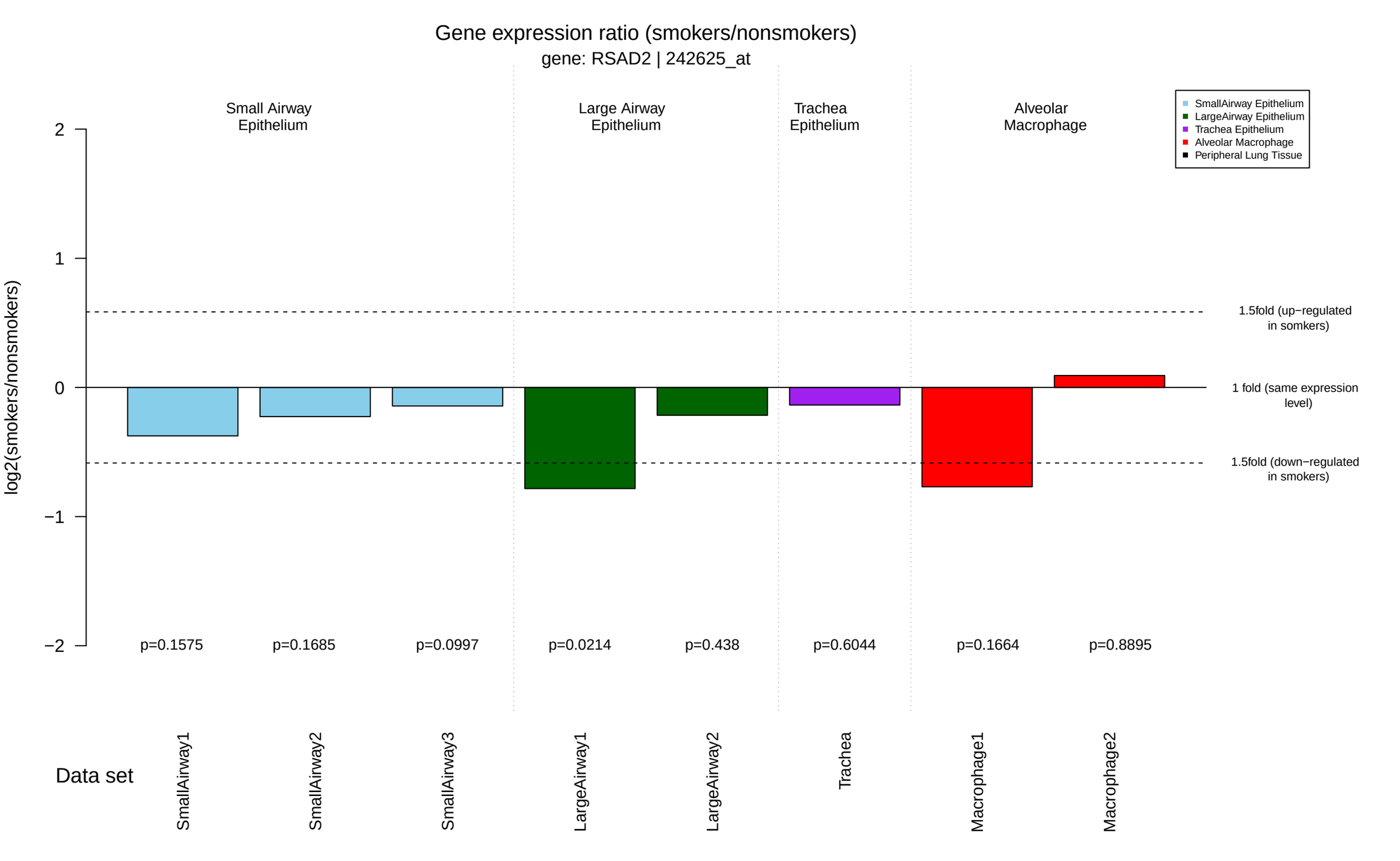
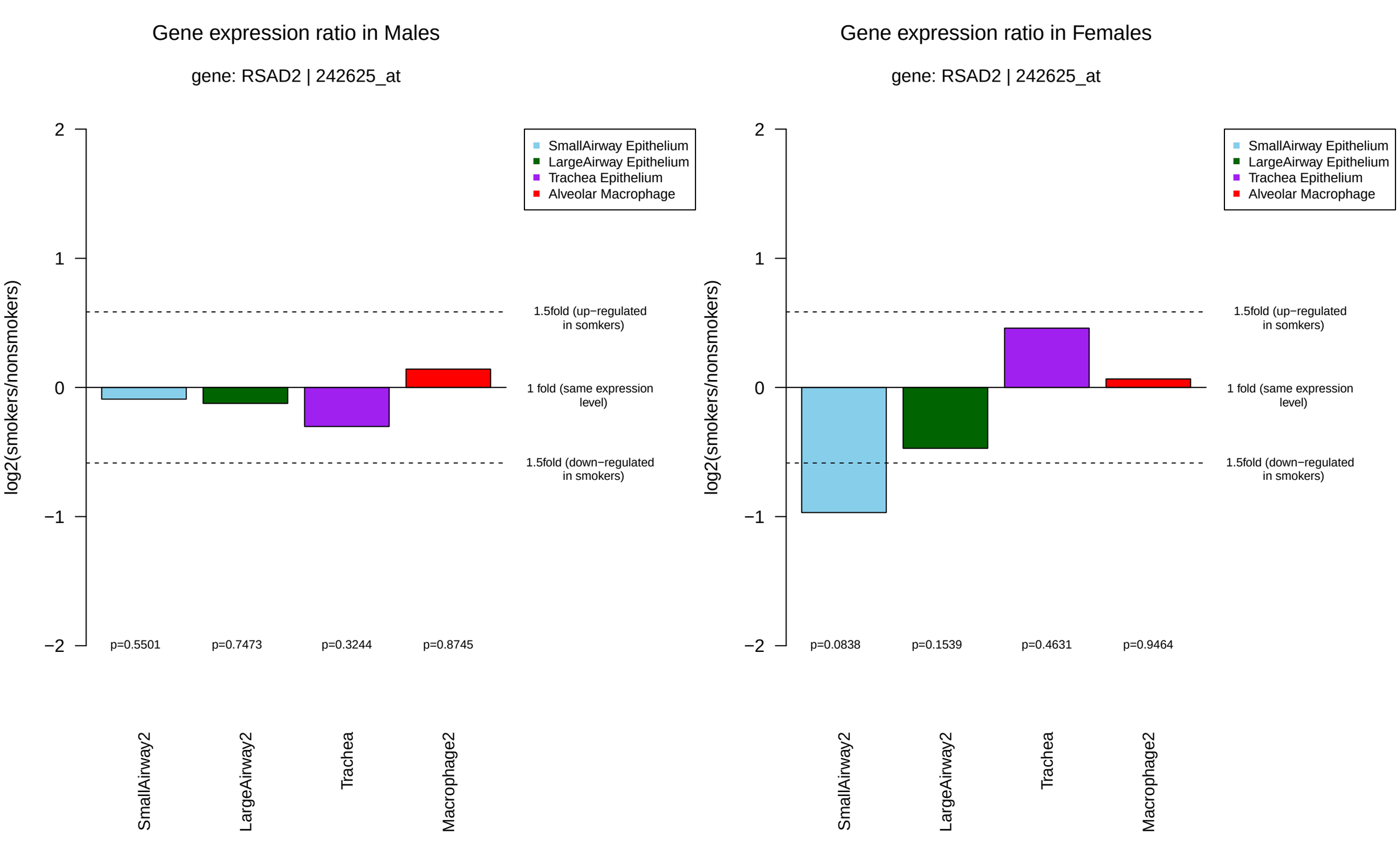
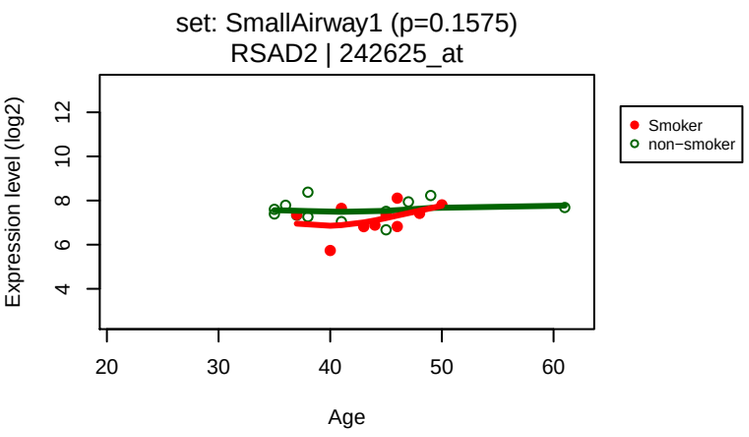
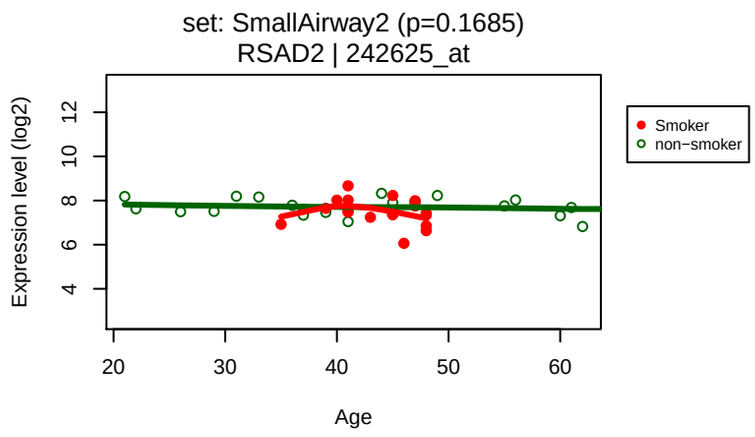
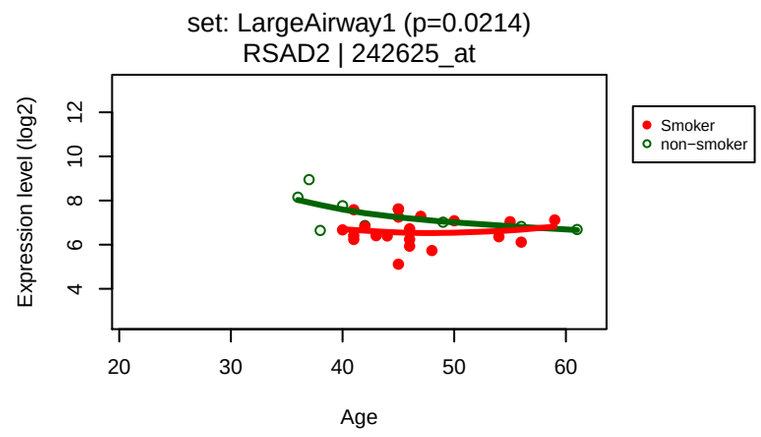

Gene expression of smokers and nonsmokers
gene: RSAD2 | 242625_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1575
p=0.1685
p=0.0997
p=0.0214
p=0.438
p=0.6044
p=0.1664
p=0.8895
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

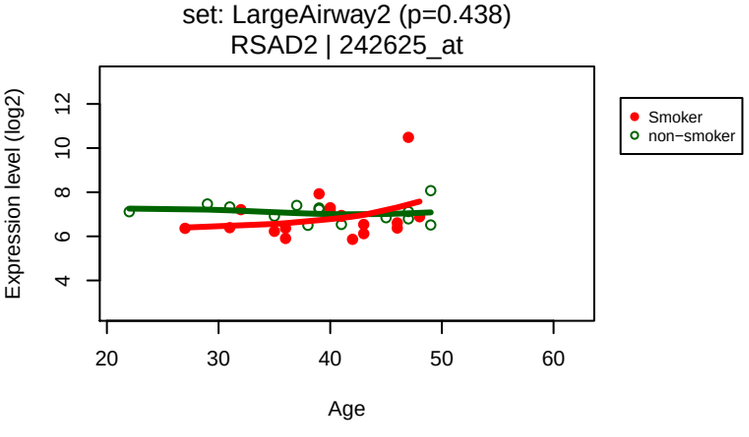

Gene expression ratio (smokers/nonsmokers)
gene: RSAD2 | 242625_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1575
p=0.1685
p=0.0997
p=0.0214
p=0.438
p=0.6044
p=0.1664
p=0.8895
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

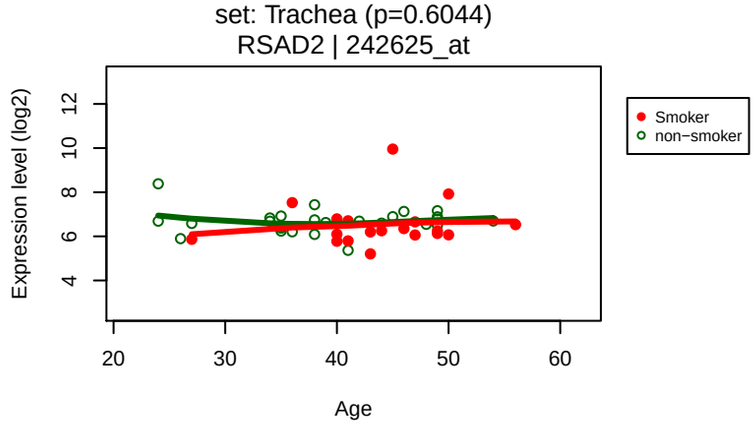

Gene expression ratio in Males
gene: RSAD2 | 242625_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5501
p=0.7473
p=0.3244
p=0.8745
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RSAD2 | 242625_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0838
p=0.1539
p=0.4631
p=0.9464
SmallAirway2
LargeAirway2
Trachea
Macrophage2

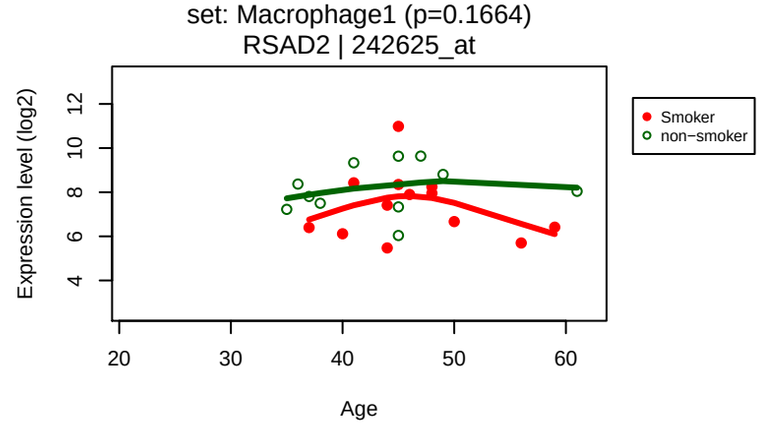

set: SmallAirway1 (p=0.1575)
RSAD2 | 242625_at
Expression level (log2)
Age
Smoker
non−smoker

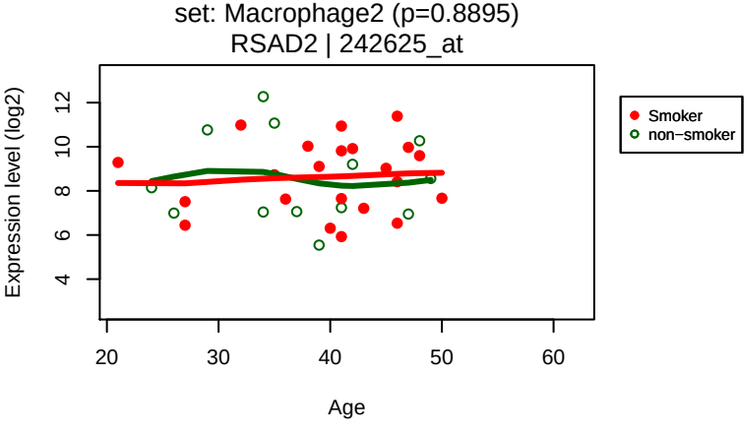

set: SmallAirway2 (p=0.1685)
RSAD2 | 242625_at
Expression level (log2)
Age
Smoker
non−smoker

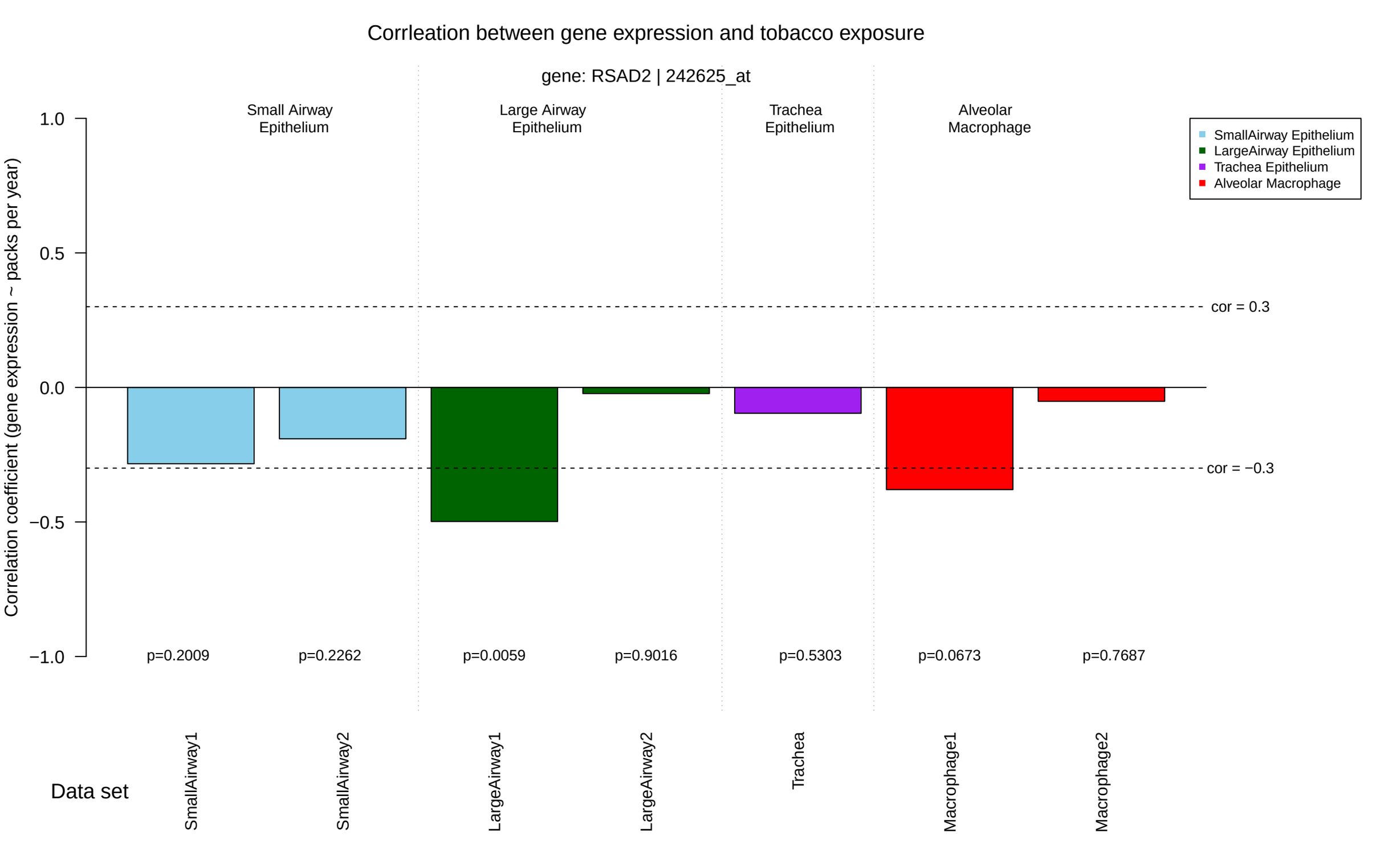

set: LargeAirway1 (p=0.0214)
RSAD2 | 242625_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.438)
RSAD2 | 242625_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6044)
RSAD2 | 242625_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1664)
RSAD2 | 242625_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8895)
RSAD2 | 242625_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RSAD2 | 242625_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2009
p=0.2262
p=0.0059
p=0.9016
p=0.5303
p=0.0673
p=0.7687
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2